

AI PROMPTS USED DURING PROJECT DEVELOPMENT

Tool: AI assistant (LLM), used for guidance and PySpark code generation.

Project: Data Management 2 final project - Databricks Medallion sales pipeline.

1. I am building a Data Management project in Databricks using a Medallion Architecture (Bronze, Silver, Gold). Help me design a pipeline that combines a Delta table and a JSON streaming source using PySpark.
2. Generate PySpark code to read a CSV dataset, create a product master table, and save it as a Delta table in Databricks Unity Catalog.
3. Generate PySpark code using Databricks Auto Loader to ingest JSON files incrementally from a Unity Catalog Volume and store them in a Bronze layer table.
4. How can I join data from a Delta table and a streaming source in Databricks using PySpark? Use StockCode as the join key and calculate revenue as $\text{Quantity} \times \text{UnitPrice}$.
5. Generate PySpark code to create a Silver layer that enriches order data with product information and prepares it for business analytics.
6. Create PySpark aggregation queries for the following business objectives:
 - Revenue by Country
 - Top 10 Products by Units Sold
7. How can I implement an incremental test in Databricks to demonstrate that newly added records are automatically processed and reflected in downstream tables?
8. Suggest a suitable Medallion Architecture explanation for a Databricks project report, including the purpose of Bronze, Silver, and Gold layers.
9. Review my PySpark code and suggest improvements for readability, documentation, and best practices in Databricks.